

WASP-163b

R-2314

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-163_250701 · created 2026-08-18T00:25:22+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460857.765 +/- 0.062 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.146 +/- 0.043
Transit depth (Rp/Rs)^2	2.13 +/- 1.26 [%]
Orbital Inclination (inc)	85.69 +/- 2.69
Airmass coefficient 1 (a1)	0.06 +/- 0.04
Airmass coefficient 2 (a2)	1.92 +/- 1.27
Scatter in the residuals of the lightcurve fit is	58.83 %
Best Comparison Star	None
Optimal Aperture	1.11
Optimal Annulus	3.0
Transit Duration (day)	0.059 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.146 ± 0.043 vs 0.119 (deviation 0.53σ)
Duration fit vs published	0.059 d vs 0.0942 d (deviation 0.96σ)
Mid-time O–C	16.2 min
Depth SNR	2.9
Residual scatter	58.83 %

Light curve

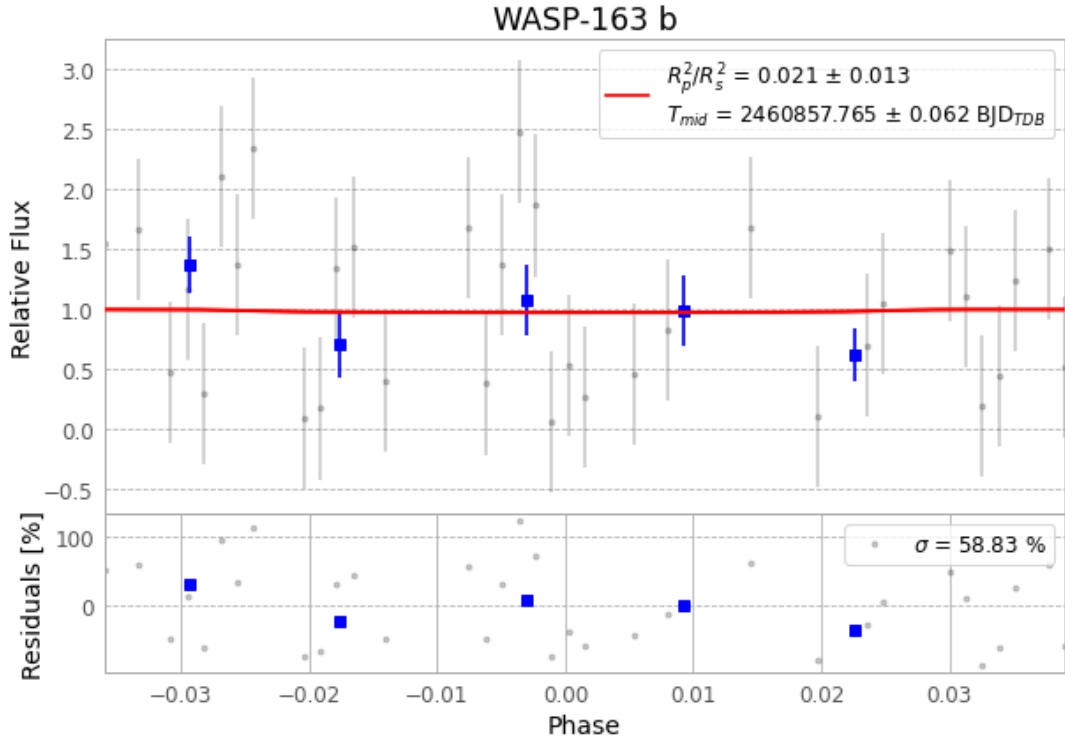


Figure 1 — FinalLightCurve_WASP-163 b_2025-06-30.png (SHA-256 2492ca5b673dddef...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [583, 162], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 01736c789f046bf7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

60 FITS frames (39 MB), WASP-163250701044816.FITS → WASP-163250701074515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-163-250701.log (a5b3aee938ec...), FinalLightCurve_WASP-163 b_2025-06-30.png (2492ca5b673d...), FinalLightCurve_WASP-163 b_2025-06-30.csv (e7ca3dd4314f...)

Compute resources

Wall time 143.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 00:25Z.